

Roberto Eisenlohr

AI Solutions Architect | Quant Researcher & Developer | Enterprise Financial Systems

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Portfolio: View Projects
Location: Manhattan, New York, NY

Summary

AI Solutions Architect with **23+ years of experience** as a professional Engineer. Building production systems at the intersection of **artificial intelligence** and **financial markets**. Recently completed an 11-month engagement at **Citigroup** where I led development of an enterprise GenAI Ecosystem for portfolio analysis — serving C-level operations across equities, options, currencies, and futures.

Core Expertise:

- **Agentic AI Systems** — LangGraph cyclic reasoning, multi-model orchestration (Gemini, Claude, Grok, Ollama), MCP server integration
- **Financial Domain** — Portfolio analysis, risk assessment, regulatory compliance, algorithmic trading
- **Full-Stack Delivery** — Architecture → Development → Deployment → Monitoring (Python, AWS, Docker, FastAPI)

Expert in building autonomous agentic systems for risk assessment, portfolio analysis, and regulatory risk review using cutting-edge Gen-AI integration, multi-agent frameworks, and large language models. Proven capability to integrate and leverage advanced technologies (LLMs, MCP Servers, AWS, GCP Vertex, Gemini, Claude, Ollama) to deliver robust, real-time solutions for quantitative teams. Skilled at dynamic Python code generation, debugging, and execution, automated HTML/CSS reporting, and end-to-end deployment of cloud-based platforms.

Credentials: Two Engineering degrees, MBA coursework in Finance & Corporate Law (FGV, IBMEC), Harvard AI Certification, SmythOS AI Agent Engineer, FINRA Series 57, 32 Professional Certifications

Experience

Citigroup / Citibank Quantitative Developer – GenAI Architect

Mar 2025 – Jan 2026

- Architected enterprise GenAI Ecosystem for portfolio analysis serving C-level operations across the institution.
- Designed and implemented agentic AI frameworks using Python and LangGraph, enabling autonomous reasoning and acting loops for complex financial workflows.
- Integrated multiple MCP (Model Context Protocol) servers for seamless tool orchestration across trading systems and data sources.

- Built intelligent agents for risk assessment, portfolio analysis, and regulatory risk review using multi-model architectures (Gemini, Claude).
- Developed, implemented, and supported large-scale platforms and agentic systems that operate across multiple departments, countries, companies, and portfolios, tailored for C-Level operations and decision-making.
- Program autonomous agents to interpret Quant queries, dynamically write, debug, and execute Python code in real time, and iteratively enhance solutions through runtime analysis.
- Develop agents capable of automatically generating and refining HTML/CSS visualizations, facilitating clearer interpretation of quantitative solutions for quants.
- Led end-to-end systems development: from ideation and initial design through programming, implementation, and deployment—including LLM model training and integration.
- Extensive hands-on experience with SQL, AWS (S3, EC2), Google Vertex AI, Gemini, Claude, building complex AI-powered platforms and agentic systems.
- Managed financial assets: Equity Options, Stocks, Credit, ETFs, Currencies, and Futures.
- Led a team of 7 high-performance Engineers throughout building the Ecosystem.
- **Saved quantitative research teams thousands of hours annually through intelligent automation of research workflows.**
- **Reduced time-to-insight for quantitative research from days to minutes through intelligent task decomposition and parallel AI agent execution via MCP protocol.**
- Collaborated with Quant Developers, Quant Analysts, Directors, Risk Analysts, Portfolio Managers, and Quant Traders, delivering transformative solutions to multi-disciplinary teams.



ABFS TECH

Quant Researcher, Developer & AI Architect

March 2016 – Present

- Developed and implemented AI-driven trading systems outperforming the S&P 500 by 3X over 10 years.
- Engineered a grid system with ROI averaging 10% monthly, Sharpe Ratio of 4.7.
- Designed and implemented AI trading systems for hedge funds, brokerage firms, family offices, proprietary traders.
- API integrations for Charles Schwab, TD Ameritrade, Interactive Brokers, IG, FOREX.com, Pepperstone.
- Developed in Python, C#, C++, NinjaScript 8, MQL5, MQL4, ThinkScript, Pine Script.
- Programmed GenAI applications—Google Gemini, Anthropic's Claude, Opus, xAI's Grok, OpenAI's ChatGPT, Ollama.
- Integrated financial platforms, built smartphone applications, connected systems within financial markets.
- Extensive expertise with AWS cloud computing solutions.
- Worked closely with Portfolio Managers, Front Office, Middle Office, Back Office, and teams from multiple Hedge Funds, Family Offices, Banks, and other financial institutions.
- **Provided quant research, development, and technology services to multiple clients across the financial markets, collaborating and delivering solutions tailored for each client's unique requirements, diverse asset classes, and business strategies.**



T3 Trading Group

Quant Researcher & Developer (Part-Time)

Feb 2019 – Dec 2019

- Developed algorithmic trading solutions for Stocks, Equity Options, ETFs using machine learning/statistics.
- Built, backtested, optimized trading algorithms and financial performance tests in Python, SQL, C#, C++.

- Led API integrations for real-time trading and data workflows.
- Partnered with Portfolio Managers, Traders, Developers to deliver production strategies.

Freelance Software Developer | Fully Remote Full-Stack Developer & Technical Consultant

July 2015 – February 2016

- Delivered custom software solutions for clients across financial services, e-commerce, and technology sectors.
- Built web applications and APIs using Python, C#, JavaScript, SQL with focus on scalable architecture and clean code.
- Developed trading tools and portfolio analytics dashboards for individual traders and small investment firms.
- Implemented database designs, RESTful APIs, and front-end interfaces for data visualization.
- Managed full project lifecycle from client requirements gathering through deployment and maintenance.

Brinpa Investments and Participations Investments Director (Part-Time)

Jan 2006 – Jun 2015

- Managed early-stage investments, following up through final payout.
- Responsible for performance analysis and investment project selection.

Key Achievements

Agentic AI Framework & MCP Infrastructure: Planned, developed, and deployed enterprise agentic system with Model Context Protocol (MCP) architecture, building custom MCP servers and clients that orchestrate multiple AI operations and saved quantitative teams thousands of hours annually by automating research workflows

Execution Optimization: Reduced latency by 95% and improved implementation shortfall by 15% for equity options flow

Risk Model Deployment: Built ML-based risk models reducing portfolio drawdown by 18% while maintaining target returns

Quantitative Research: Developed systematic trading strategy with Sharpe Ratio of 4.7 and 10% average monthly returns over multi-year live trading period

3X S&P 500 Outperformance: AI-driven trading systems outperforming benchmark over 10-year backtest with rigorous walk-forward validation

Real-Time Risk Monitoring: Implemented production risk system alerting on portfolio breaches within 100ms of market data updates

Skills

Programming Languages

Python C# C++ SQL HTML CSS JavaScript MQL5 MQL4 NinjaScript 8 ThinkScript Pine Script

Software & Tools

Amazon Web Services (AWS) Google Gemini Anthropic's Claude Opus Sonnet OpenAI's ChatGPT Ollama Metatrader Charles Schwab Trading Platform
TradingView Alveo C# Platform Unity

Quantitative Analysis

Algorithmic Trading Quantitative Research Mathematical Modeling Statistical Analysis Risk Management Portfolio Optimization Backtesting
Derivatives Pricing Monte Carlo Simulation Time Series Analysis Financial Algorithms Options & Futures ETFs & Indexes Pearson Correlation Analysis

Trading Strategies

Statistical Arbitrage Pairs Trading Mean Reversion Momentum Market Making Execution Algorithms Grid Trading Trend Following Target Achievement
Triangular Arbitrage Dynamic Circular Arbitrage 300+ Strategies Developed

Risk Management

VaR (Value at Risk) CVaR (Conditional Value at Risk) Greeks Analysis (Delta, Gamma, Theta, Vega) Scenario Analysis Stress Testing Drawdown Control
Position Sizing

Artificial Intelligence (AI)

Machine Learning AI Trading Systems Predictive Modeling Neural Networks Model Optimization Deep Learning Natural Language Processing
Large Language Models (LLM) Generative AI (GenAI) Agentic AI LangGraph LangChain LangSmith Model Context Protocol (MCP)

Financial Expertise

Financial Engineering Quantitative Finance Financial Modeling Cryptocurrency Trading Forex Trading Equities CFDs Portfolio Management
Financial Analysis Equity Options Stocks Futures Futures Options Credit Instruments Currencies

Other Skills

API Integration Mobile App Development Project Management Client Communication Strong Analytical & Research Skills Problem-solving
Excellent Written & Verbal Communication Low Latency Development Docker FastAPI Playwright

Education

Harvard University

Certification, Artificial Intelligence in Python

August 2023 – January 2024

Federal Fluminense University (UFF)

1st Engineering Degree

Diploma, 2002

UNESA

2nd Engineering Degree

Diploma, 2014

Ibmec

Attended 50 credits towards MBA, Finance

2005–2006

FGV – Getulio Vargas Foundation

Attended 36 credits towards MBA, Corporate Law

2003–2004

Certifications & Licenses

AI & Machine Learning (7):

Beyond Vibe Coding with Google Gemini (LinkedIn Learning, Dec 2025)

AI Agent Engineer (SmythOS, Sep 2024)

Harvard AI in Python (Harvard University, Jan 2024)

Prompt Engineering for ChatGPT (Vanderbilt, Feb 2024)

Artificial Intelligence (Google Launchpad)

Machine Learning (Google Launchpad)

Data Science (Google Launchpad)

AWS Certifications (6):

- API Serverless Gateway
- AWS Cloud Computing
- Decoupled Applications
- Deploying & Scaling with Lightsail
- Lambda Foundations
- MySQL

Google Launchpad (18):

- Artificial Intelligence
- Blockchain
- Cybersecurity
- Data Science
- Docker
- Firebase
- GIT
- HoneyCode
- Machine Learning
- Node.js
- Programming in C#
- Python 3
- Python Advanced
- SQL Programming
- Social Media Marketing
- Solidity/Ethereum Blockchain
- Unity Programming
- Website Building

Other (4):

- Licensed Engineer (Crea-RJ)
- PHP Programming (Sololearn)
- Series 57 - Securities Trader Representative Exam (FINRA, previously held)
- WooCommerce Themes

Total: 32 Professional Certifications

Publications

Cryptocurrency – Starter's Guide for Traders and Investors

ISBN: 978-1-63901-260-2

Publisher: Primedia

Publication Date: May 2021

Available at: <https://cryptocurrency-book.com/>